BCB5460 R_Assignment

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

File Inspection

fang_et_al_genotypes.txt

```r
# load dataframe using relative path
fang <- read.csv("Input_files/fang_et_al_genotypes.txt", header = FALSE, sep = "\t")

# it has 2782 rows and 986 columns (row number excluding header)
dim(fang)
```

```
## [1] 2783  986
```

```r
# header contains "Sample_ID", "JG_OTU", "Group", and all SNP names
col <- fang[1,]
col[1:10]
```

```
##          V1     V2    V3       V4       V5    V6    V7    V8    V9   V10
## 1 Sample_ID JG_OTU Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6
```

```r
# file size
file.size("Input_files/fang_et_al_genotypes.txt")
```

```
## [1] 11051939
```

snp_position.txt

```r
# load dataframe using relative path
snp <- read.csv("Input_files/snp_position.txt", header = FALSE, sep = "\t")

# it has 983 rows and 15 columns (row number excluding header)
dim(snp)
```

```
## [1] 984  15
```

```r
# header contains "SNP_ID", "Chromosome", Position", and more shown in standard out
col <- snp[1,]
col[1:10]
```

```
##       V1            V2         V3       V4      V5             V6       V7
## 1 SNP_ID cdv_marker_id Chromosome Position alt_pos mult_positions amplicon
##                      V8   V9              V10
## 1 cdv_map_feature.name gene candidate/random
```

```r
# file size
file.size("Input_files/snp_position.txt")
```

```
## [1] 82763
```

Data Processing

extract maize and teosinte data from `fang`; change the `Sample_ID` header to `SNP_ID`; delete columns "JG_OTU" and "Group"

```r
maize <- fang %>% filter(.[[3]] %in% c("ZMMIL", "ZMMLR", "ZMMMR", "Group"))
teosinte <- fang %>% filter(.[[3]] %in% c("ZMPBA", "ZMPIL", "ZMPJA", "Group"))

maize[1,1] <- "SNP_ID"
teosinte[1,1] <- "SNP_ID"

maize <- maize %>% select(-c(2,3))
teosinte <- teosinte %>% select(-c(2,3))
```

transpose maize and teosinte data

```r
tp_maize <- as.data.frame(t(maize))
tp_teosinte <- as.data.frame(t(teosinte))
```

trim `snp` so that it only has columns "SNP_ID", "Chromosome", and "Position"

```r
trim_snp <- snp %>% select(c(1,3,4))
```

merge `trim_snp` with `tp_maize` and `tp_teosinte`; note that the header is no longer at the top. Save the header as an object to add back to the output files in the end because it is going to be lost in following steps

```r
join_maize <- merge(trim_snp,tp_maize, by.x = 1, by.y = 1)
join_teosinte <- merge(trim_snp,tp_teosinte, by.x = 1, by.y = 1)

header1 <- join_maize %>% filter(.[[1]] %in% c("SNP_ID"))
header2 <- join_teosinte %>% filter(.[[1]] %in% c("SNP_ID"))
```

make three files from `join_maize` with values for position, unknown positions (up), and multiple positions (mp). Header is add back to "maize_up" and "maize_mp"

```
SeparatePosition <- function(file) {
  file_p <- file %>% filter(!.[[2]] %in% c("unknown", "multiple")
                                & !.[[3]] %in% c("unknown", "multiple","Position"))
  file_up <- file %>% filter(.[[2]] %in% c("unknown") | .[[3]] %in% c("unknown" ))
  file_mp <- file %>% filter(.[[2]] %in% c("multiple") | .[[3]] %in% c("multiple"))
  return(list(file_p, file_up, file_mp))
}

results <- SeparatePosition(join_maize)
maize_p <- as.data.frame(results[1])
maize_up <- as.data.frame(results[2])
maize_mp <- as.data.frame(results[3])

colnames(maize_up) <- header1
colnames(maize_mp) <- header1

write.table(maize_up, file = paste0("Output_files/maize_up.csv"), sep = "\t", row.names = FALSE, quote =
write.table(maize_mp, file = paste0("Output_files/maize_mp.csv"), sep = "\t", row.names = FALSE, quote =
```

Separate `maize_p` with chromosome number and sort in increasing and decreasing order (Note: it is sorted numerically but not in a mathematically correct way, e.g. 9300391 > 88863284 because 9>8 regardless of the number of digits). Replace "?" to "-" in the decreasing-order files. Header is added back to the files in the "OutputFile()" function

```
Separate_Sort <- function(file) {
  # separate with chr number
  chr <- list()
  for (i in 1:10) {
    file_i <- file %>% filter(.[[2]] %in% c(i))
    chr[[i]] <- file_i
  }
  # sort in increasing order
  inc <- list()
  for (i in 1:10) {
    i_sort <- chr[[i]] %>% arrange(as.numeric(.[[3]]))
    inc[[i]] <- i_sort
  }
  # sort in descending order and replace ? with -
  dec <- list()
  for (i in 1:10) {
    d_sort <- chr[[i]] %>% arrange(desc(as.numeric(.[[3]])))
    d_sort[d_sort == "?/?"] <- "-/-"
    dec[[i]] <- d_sort
  }
  output <- list(inc,dec)
  return(output)
}


OutputFile <- function(file) {
  results <- Separate_Sort(file)
  inc_file <- results[[1]]
  for (i in 1:10) {
```

```r
    chri <- as.data.frame(inc_file[i])
    # maize and teosinte data have different number of columns, so different header
    if (species == "maize") {
      colnames(chri) <- header1
    } else {
      colnames(chri) <- header2
    }

    filename <- paste0(species,c("_chr"),i,c("_in.csv"))
    write.table(chri, file = paste0("Output_files/",filename), sep = "\t", row.names = FALSE, quote = FA
  }

  dec_file <- results[[2]]
  for (i in 1:10) {
    chri <- as.data.frame(dec_file[i])
    # again, different header for maize and teosinte
    if (species == "maize") {
      colnames(chri) <- header1
    } else {
      colnames(chri) <- header2
    }

    filename <- paste0(species,c("_chr"),i,c("_de.csv"))
    write.table(chri, file = paste0("Output_files/",filename), sep = "\t", row.names = FALSE, quote = FA
  }
}

species <- "maize"
OutputFile(maize_p)
```

For teosinte data, repeat the process of making files that have position value and have unknown or multiple positions, separating by chromosome number, sorting based on position value, replace "?" with "-", and output results to directory

```r
results <- SeparatePosition(join_teosinte)
teosinte_p <- as.data.frame(results[1])
teosinte_up <- as.data.frame(results[2])
teosinte_mp <- as.data.frame(results[3])

colnames(teosinte_up) <- header2
colnames(teosinte_mp) <- header2

write.table(teosinte_up, file = paste0("Output_files/teosinte_up.csv"),
            sep = "\t", row.names = FALSE, quote = FALSE)
write.table(teosinte_mp, file = paste0("Output_files/teosinte_mp.csv"),
            sep = "\t", row.names = FALSE, quote = FALSE)

species <- "teosinte"
OutputFile(teosinte_p)

# check one file as a representative to see if the files created are tab-delimited
data <- read_delim("Output_files/maize_mp.csv", delim = NULL, show_col_types = FALSE)
colnames(data) %>% .[1:10]
```

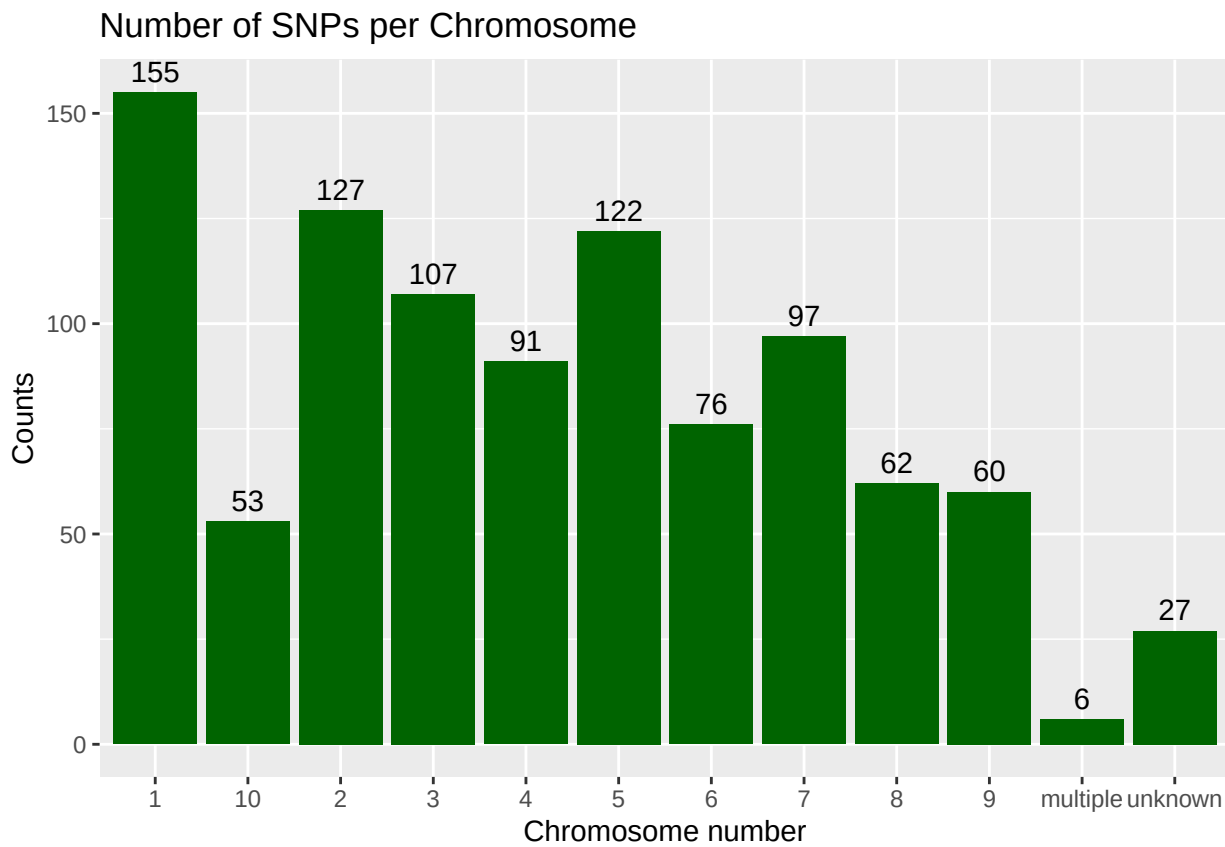
```
##  [1] "SNP_ID"     "Chromosome" "Position"   "ZDP_0752a"  "ZDP_0793a"
##  [6] "ZDP_0612a"  "ZDP_0602a"  "ZDP_0581a"  "ZDP_0552a"  "ZDP_0543a"
```

Visualization

Distribution of SNPs on and across chromosomes. Note that this is the total number of SNP positions that were investigated and not the actual number of SNPs.

```
NofSNP <- join_maize %>% filter(!.[[2]] %in% c("Chromosome")) %>%
  ggplot(aes(x=.[[2]])) + geom_bar(fill="darkgreen") + labs(title="Number of SNPs per Chromosome",x="Chr

plot(NofSNP)
```

```
## Warning: Use of `.[[2]]` is discouraged.
## i Use `.data[[2]]` instead.
## Use of `.[[2]]` is discouraged.
## i Use `.data[[2]]` instead.
```




```
ggsave("Graphs/NofSNP.png",NofSNP)
```

```
## Saving 6.5 x 4.5 in image
```

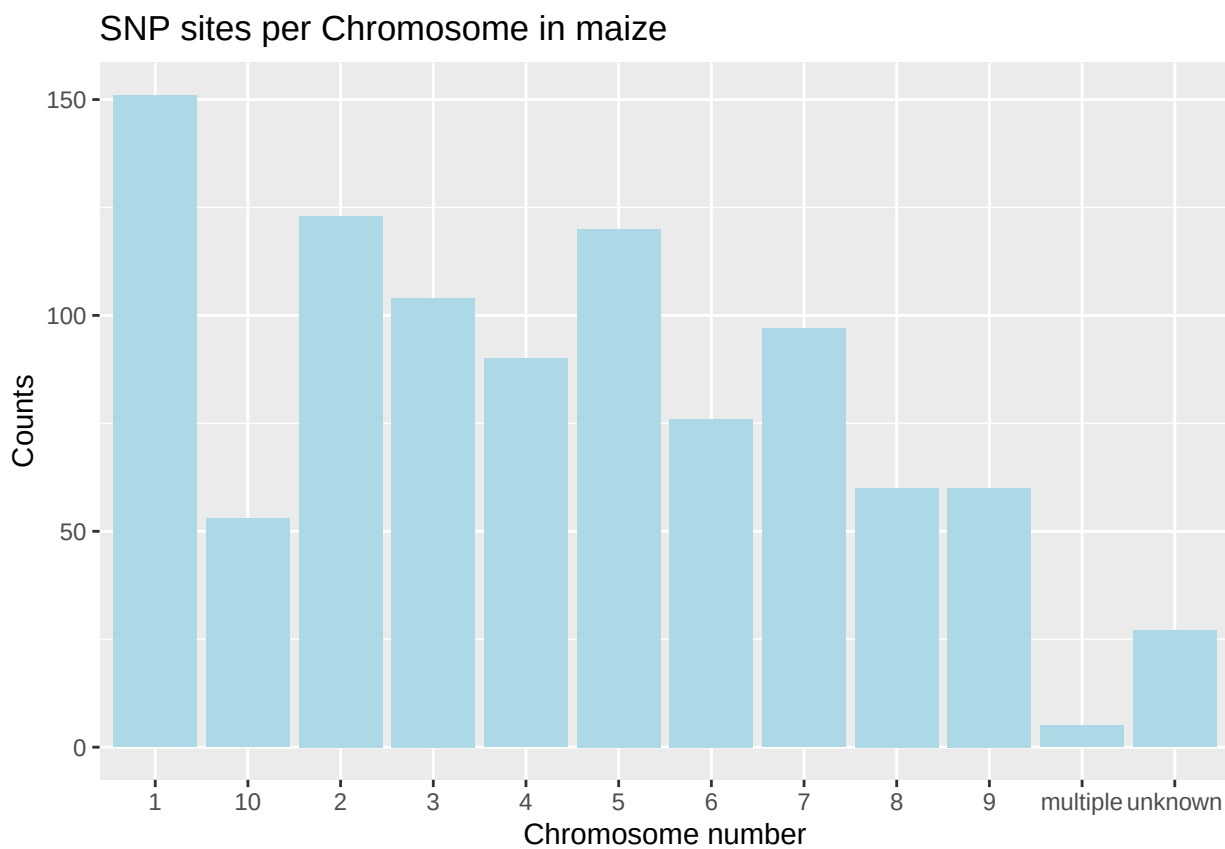
```
## Warning: Use of `.[[2]]` is discouraged.
```

```
## i Use `.data[[2]]` instead.
## Use of `.[[2]]` is discouraged.
## i Use `.data[[2]]` instead.
```

Compare SNPs in maize and teosinte.

```
# create a new data frame by copying the original and removing header, so all changes made in visualiza
join_maize_visual <- join_maize %>% filter(!.[[2]] %in% c("Chromosome"))
join_maize_visual[join_maize_visual == "?/?"] <- NA

df <- join_maize_visual %>%
  rowwise() %>%
  mutate(is_SNP = n_distinct(c_across(4:ncol(join_maize_visual)), na.rm = TRUE) == 1) %>%
  ungroup()

maize_true_snp <- df %>% filter(is_SNP %in% c("FALSE")) %>%
  count(.[[2]]) %>% ggplot(aes(x=factor(.[[1]]),y=n)) + geom_col(fill="lightblue") + labs(title="SNP si

plot(maize_true_snp)
```

```
## Warning: Use of `.[[1]]` is discouraged.
## i Use `.data[[1]]` instead.
```

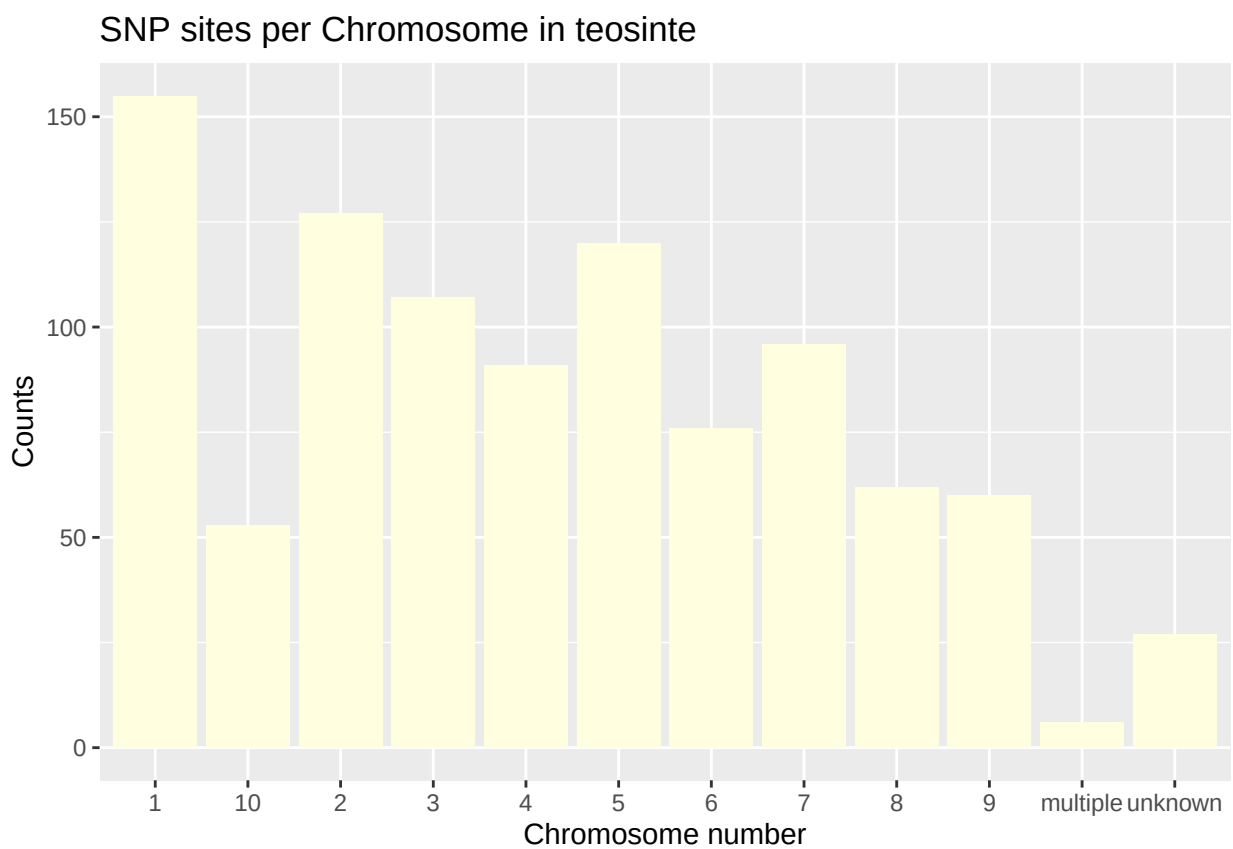


```r
ggsave("Graphs/maize_true_snp.png",maize_true_snp)
```

```
## Saving 6.5 x 4.5 in image

## Warning: Use of `.[[1]]` is discouraged.
## i Use `.data[[1]]` instead.
```

```r
# repeat for teosinte
join_teosinte_visual <- join_teosinte %>% filter(!.[[2]] %in% c("Chromosome"))
join_teosinte_visual[join_teosinte_visual == "?/?"] <- NA

df <- join_teosinte_visual %>%
  rowwise() %>%
  mutate(is_SNP = n_distinct(c_across(4:ncol(join_teosinte_visual)), na.rm = TRUE) == 1) %>%
  ungroup()

teosinte_true_snp <- df %>% filter(is_SNP %in% c("FALSE")) %>%
  count(.[[2]]) %>% ggplot(aes(x=factor(.[[1]]),y=n)) + geom_col(fill="lightyellow") + labs(title="SNP s

plot(teosinte_true_snp)
```

```
## Warning: Use of `.[[1]]` is discouraged.
## i Use `.data[[1]]` instead.
```



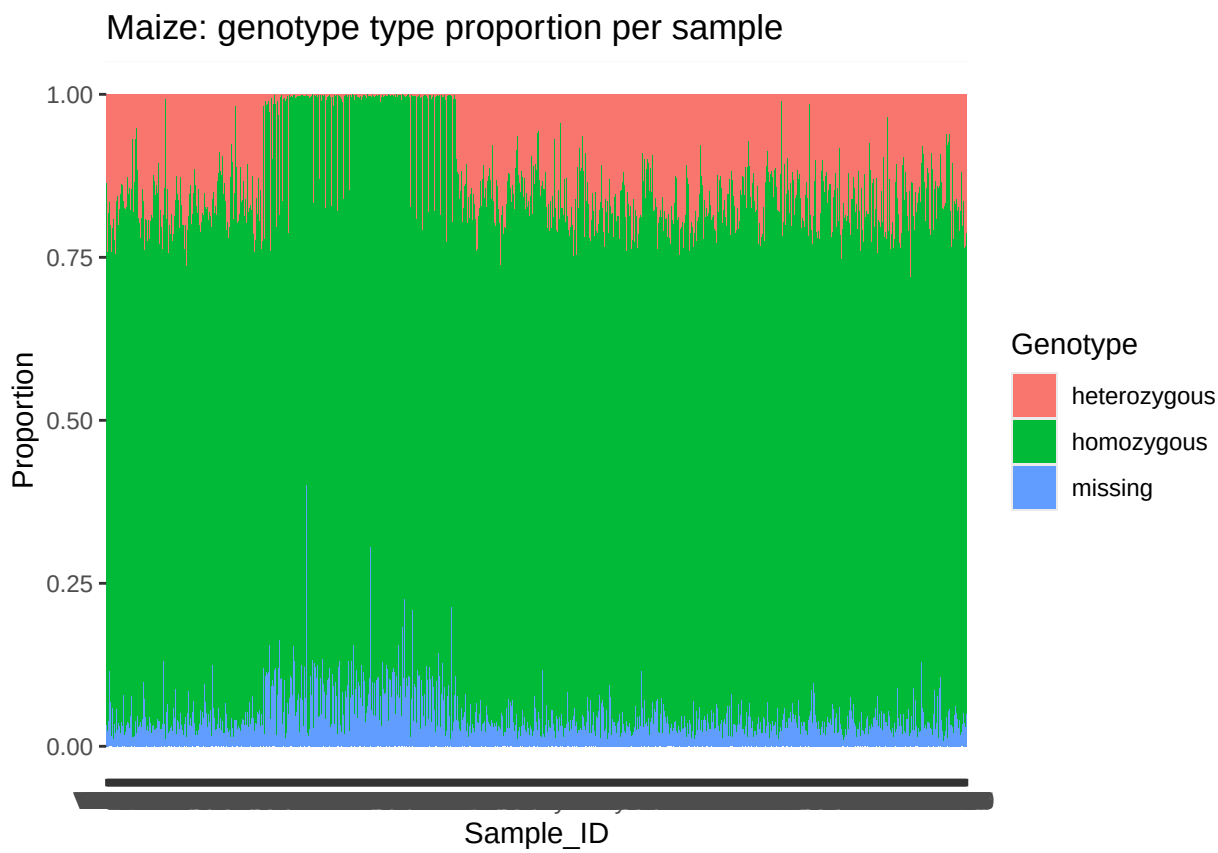
```r
ggsave("Graphs/teosinte_true_snp.png",teosinte_true_snp)
```

```
## Saving 6.5 x 4.5 in image
```

```
## Warning: Use of `.[[1]]` is discouraged.
## i Use `.data[[1]]` instead.
```

Missing data and amount of heterozygosity

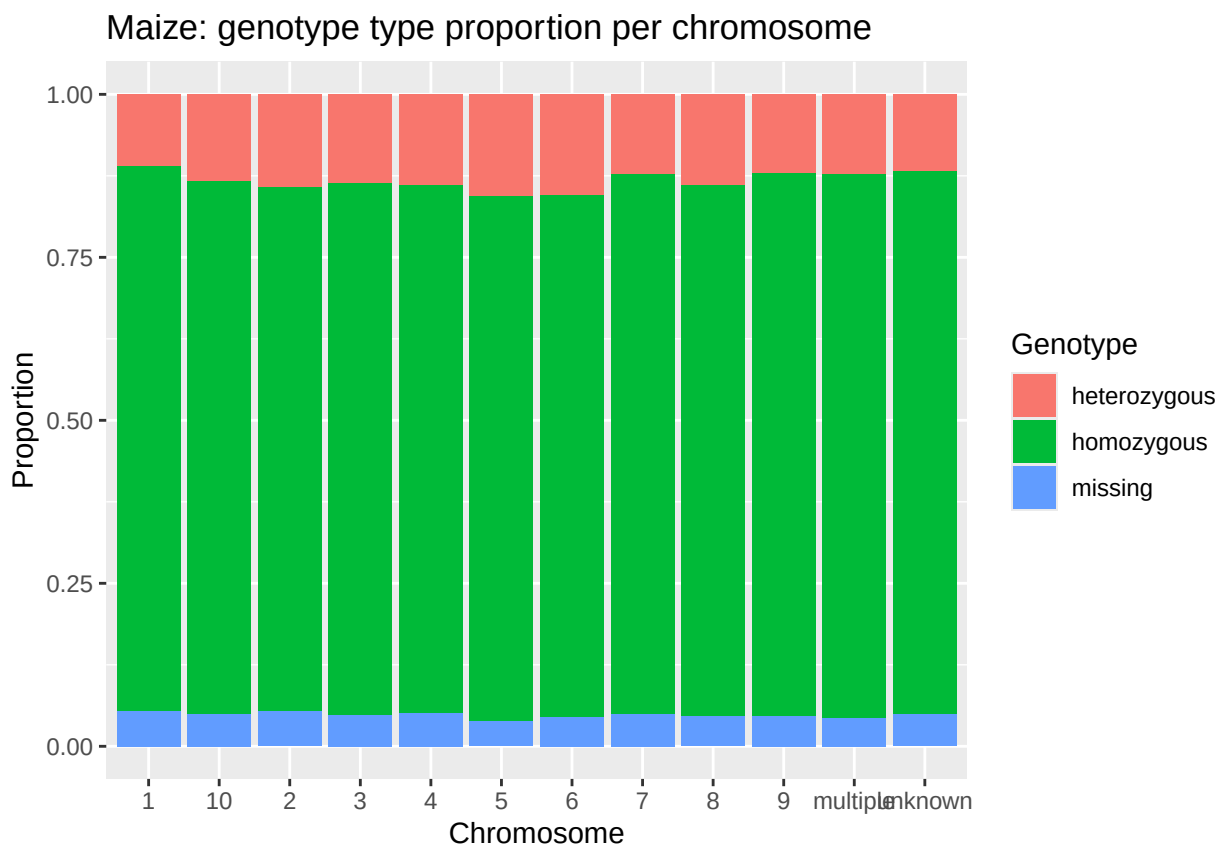
```r
# convert to long format
long_maize <- join_maize_visual %>% pivot_longer(cols = 4:ncol(join_maize_visual),
               names_to = "sample_ID",
               values_to = "genotype")

# classify genotype type to missing, homozygous, and heterozygous
long_maize <- long_maize %>%
  mutate(genotype_type = case_when(is.na(genotype) ~ "missing",genotype %in%
                                     c("A/A","T/T","C/C","G/G") ~ "homozygous",
                                   TRUE ~ "heterozygous"))

# calculate proportion of genotype types in each sample
genotype_prop <- long_maize %>%
  group_by(sample_ID, genotype_type) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(sample_ID) %>%
  mutate(prop = n / sum(n))

# plot genotype type proportion per sample
type_sample_maize <- ggplot(genotype_prop, aes(x = sample_ID, y = prop, fill = genotype_type)) +
  geom_col() +
  labs(title = "Maize: genotype type proportion per sample", x = "Sample_ID",
       y = "Proportion",
       fill = "Genotype")
plot(type_sample_maize)
```




```
ggsave("Graphs/type_sample_maize.png",type_sample_maize)
```

```
## Saving 6.5 x 4.5 in image
```

```
# plot genotype type proportion per chromosome across samples
colnames(long_maize)[2] <- "chromosome"
genotype_prop2 <- long_maize %>%
  group_by(chromosome, genotype_type) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(chromosome) %>%
  mutate(prop2 = n / sum(n))

type_chr_maize <- ggplot(genotype_prop2, aes(x = chromosome, y = prop2, fill = genotype_type)) +
  geom_col() +
  labs(title = "Maize: genotype type proportion per chromosome",x = "Chromosome",
       y = "Proportion",
       fill = "Genotype")
plot(type_chr_maize)
```



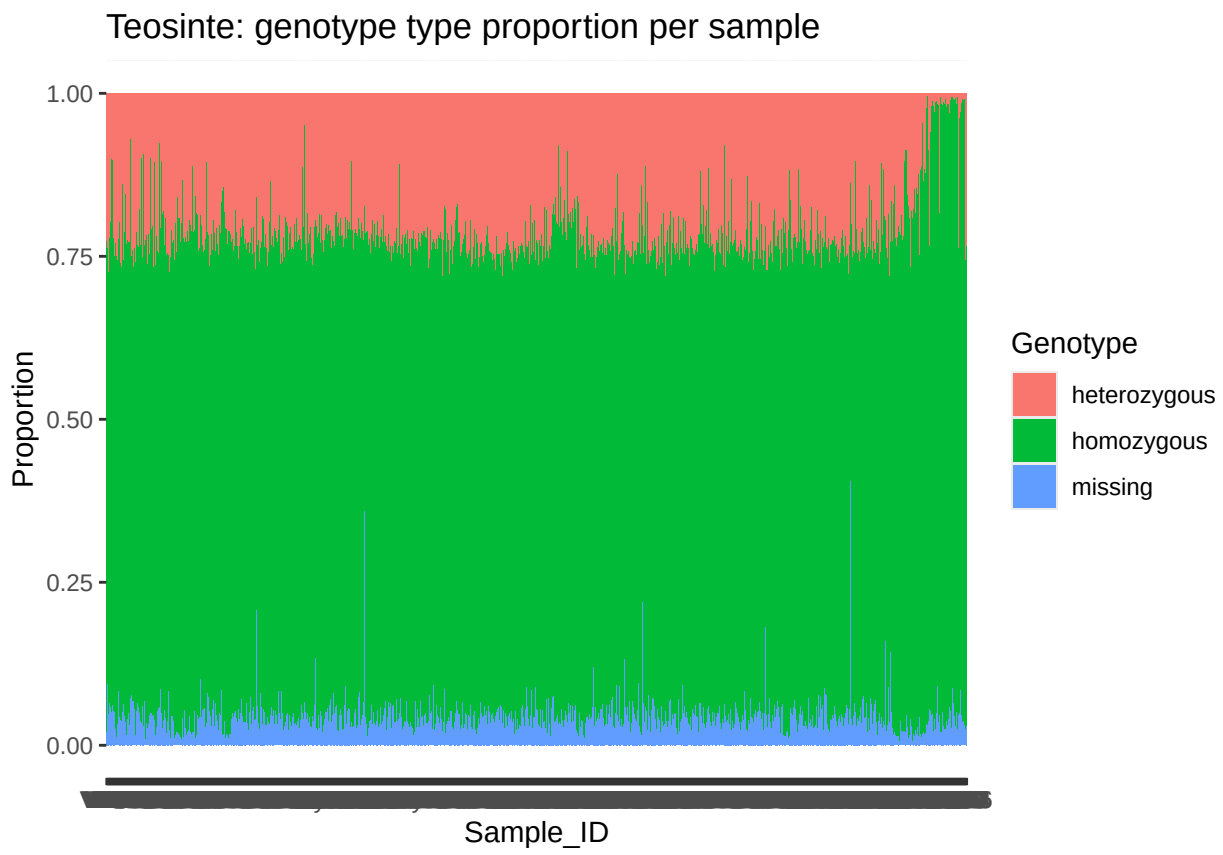

```
ggsave("Graphs/type_chr_maize.png",type_chr_maize)
```

```
## Saving 6.5 x 4.5 in image
```

Repeat for teosinte

```
# convert to long format
long_teosinte <- join_teosinte_visual %>% pivot_longer(cols = 4:ncol(join_teosinte_visual),
                names_to = "sample_ID",
                values_to = "genotype")

# classify genotype type to missing, homozygous, and heterozygous
long_teosinte <- long_teosinte %>%
  mutate(genotype_type = case_when(is.na(genotype) ~ "missing",genotype %in%
                                     c("A/A","T/T","C/C","G/G") ~ "homozygous",
                                   TRUE ~ "heterozygous"))

# calculate proportion of genotype types in each sample
genotype_prop <- long_teosinte %>%
  group_by(sample_ID, genotype_type) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(sample_ID) %>%
  mutate(prop = n / sum(n))

# plot genotype type proportion per sample
```

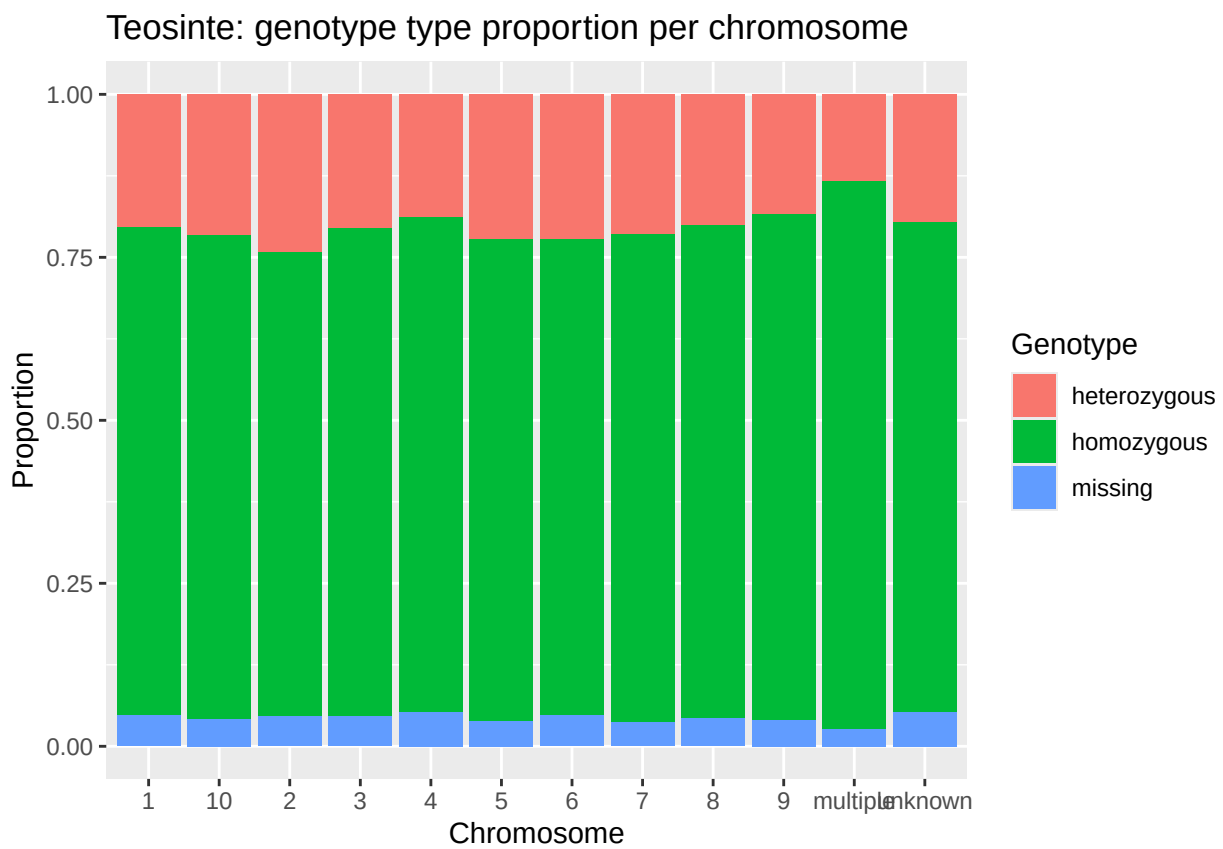
```r
type_sample_teosinte <- ggplot(genotype_prop, aes(x = sample_ID, y = prop,
                                                  fill = genotype_type)) + geom_col() +
  labs(title = "Teosinte: genotype type proportion per sample", x = "Sample_ID",
       y = "Proportion",
       fill = "Genotype")
plot(type_sample_teosinte)
```




```r
ggsave("Graphs/type_sample_teosinte.png",type_sample_teosinte)
```

```
## Saving 6.5 x 4.5 in image
```

```r
# plot genotype type proportion per chromosome across samples
colnames(long_teosinte)[2] <- "chromosome"
genotype_prop2 <- long_teosinte %>%
  group_by(chromosome, genotype_type) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(chromosome) %>%
  mutate(prop2 = n / sum(n))

type_chr_teosinte <- ggplot(genotype_prop2, aes(x = chromosome, y = prop2,
                                                fill = genotype_type)) + geom_col() +
  labs(title = "Teosinte: genotype type proportion per chromosome",x = "Chromosome",
       y = "Proportion",
       fill = "Genotype")
plot(type_chr_teosinte)
```




```
ggsave("Graphs/type_chr_teosinte.png",type_chr_teosinte)
```

```
## Saving 6.5 x 4.5 in image
```

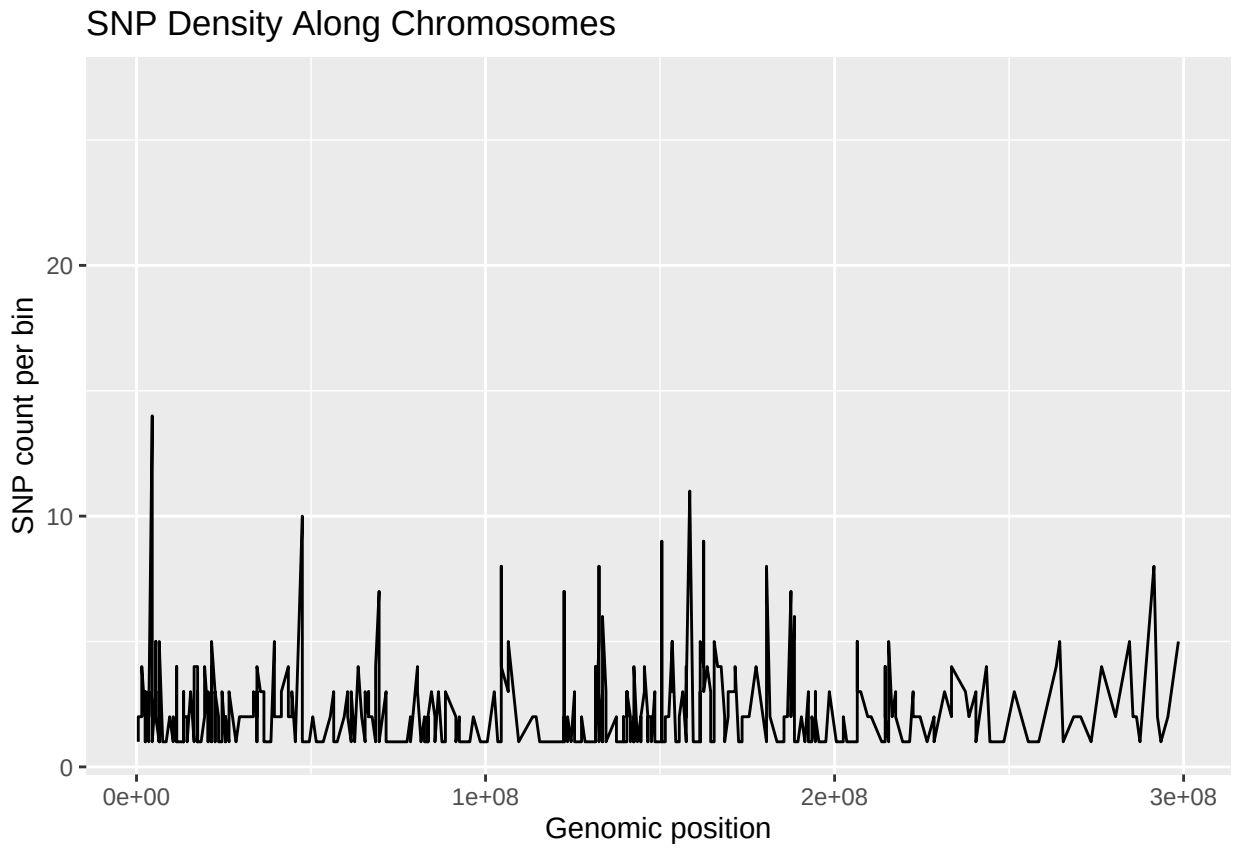
SNP density along chromosome

```
bin_size <- 1e6

density_maize <- join_maize_visual %>%
  mutate(CHR = as.factor(.[[2]]), bin = floor(as.numeric(.[[3]]) / bin_size)) %>%
  group_by(CHR, bin) %>%
  summarise(snp_count = n(), .groups = "drop") %>%
  mutate(bin_midpoint = (bin + 0.5) * bin_size)
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `bin = floor(as.numeric(.[[3]])/bin_size)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```
density_maize_plot <- ggplot(density_maize, aes(x = bin_midpoint, y = snp_count)) +
  geom_line() +
  labs(x = "Genomic position", y = "SNP count per bin", title = "SNP Density Along Chromosomes")

plot(density_maize_plot)
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```
ggsave("Graphs/density_maize_plot.png",density_maize_plot)
```

```
## Saving 6.5 x 4.5 in image
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

```
save.image(file="Sunday.RData")
```